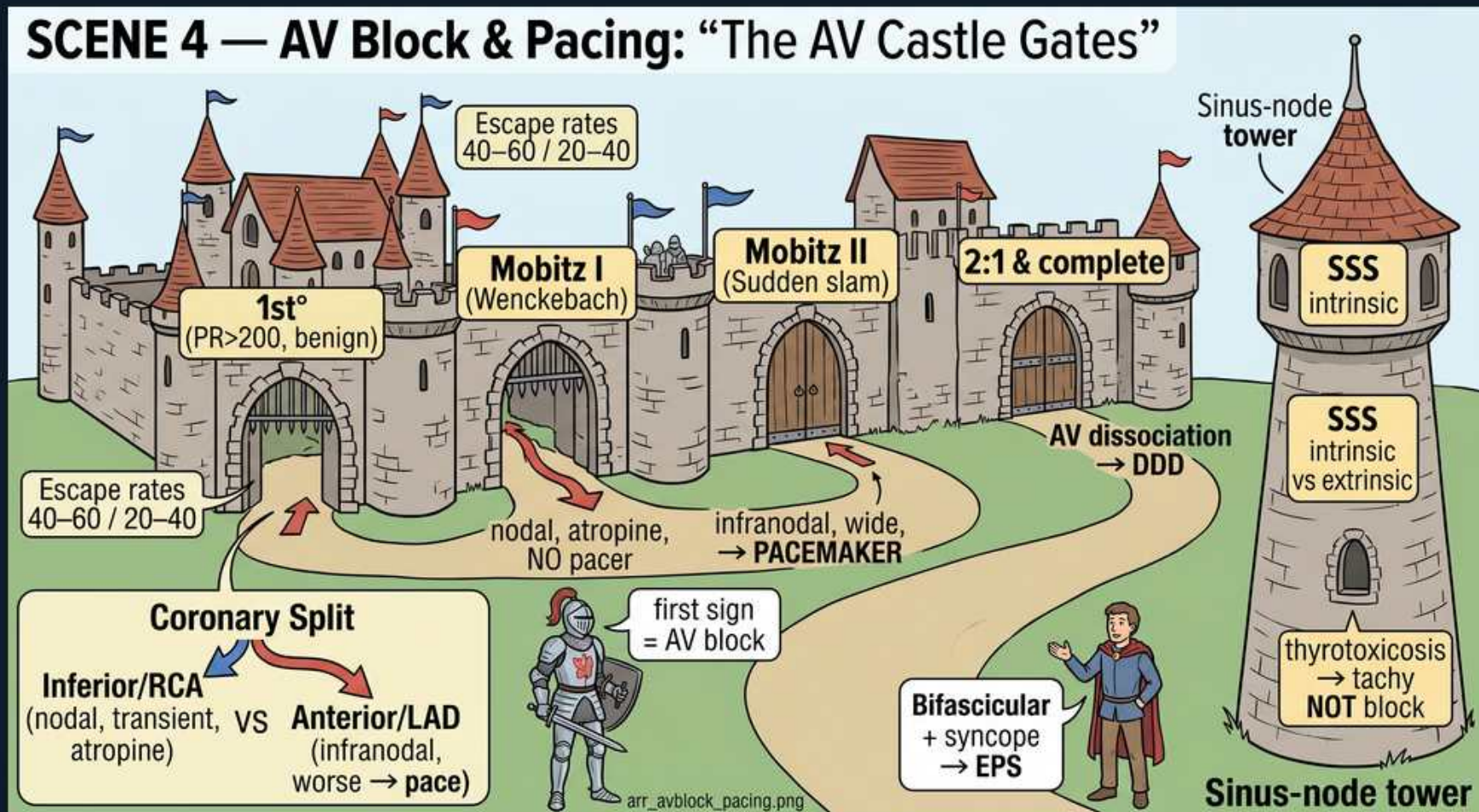


SCENE 4 — AV Block & Pacing: “The AV Castle Gates”



1. 1st-degree block

WHAT YOU SEE

Gate '1st° (PR>200, benign)'.

WHAT IT ENCODES

First-degree AV block = fixed PR >200 ms with every P conducted. Usually benign and needs no treatment; it reflects slowed AV-nodal conduction.

BOARD TRAP

WRONG: 'first-degree block needs a pacemaker.' It is benign and observed unless markedly prolonged with symptoms.

2. Mobitz I (Wenckebach)

WHAT YOU SEE

Gate 'Mobitz I (Wenckebach)' — nodal, atropine, NO pacer.

WHAT IT ENCODES

Mobitz I = progressive PR lengthening until a beat drops; the block is in the AV node, the QRS is narrow, and it usually responds to atropine. Generally benign — no pacemaker unless symptomatic.

BOARD TRAP

WRONG: 'Wenckebach needs permanent pacing.' Nodal Mobitz I is usually benign and atropine-responsive; pacing is for Mobitz II/high-grade/symptomatic block.

3. Mobitz II (sudden slam)

WHAT YOU SEE

Gate 'Mobitz II' — infranodal, wide, → PACEMAKER.

WHAT IT ENCODES

Mobitz II = sudden non-conducted P waves with a CONSTANT PR; the block is infranodal (His-Purkinje), the QRS is often wide, it does NOT respond to atropine, and it progresses to complete block — permanent pacemaker indicated.

BOARD TRAP

WRONG: 'give atropine and observe Mobitz II.' Infranodal block can worsen with atropine; it needs pacing.

4. 2:1 & complete block → DDD

WHAT YOU SEE

Gate '2:1 & complete' with AV dissociation → DDD.

WHAT IT ENCODES

Third-degree (complete) AV block = full AV dissociation with an escape rhythm; symptomatic or infranodal high-grade block requires a permanent pacemaker (dual-chamber DDD when sinus rhythm is intact to preserve AV synchrony).

BOARD TRAP

WRONG: 'complete heart block can be managed with medication.' Definitive therapy is permanent pacing.

5. Escape rates 40–60 / 20–40

WHAT YOU SEE

Banner: escape rates 40-60 / 20-40.

WHAT IT ENCODES

Escape rhythms localize the block: a junctional escape (40–60 bpm, narrow QRS) sits above/at the His; a ventricular escape (20–40 bpm, wide QRS) is infranodal and unreliable — favoring pacing.

BOARD TRAP

WRONG: 'a narrow-complex escape at 50 is dangerous and needs urgent pacing.' Junctional escapes are relatively stable; wide slow ventricular escapes are the worrying ones.

6. Sinus-node tower — SSS intrinsic vs extrinsic

WHAT YOU SEE

Tower: 'SSS intrinsic vs extrinsic'.

WHAT IT ENCODES

Sick sinus syndrome (sinus pauses, brady-tachy) is intrinsic (fibrosis/age) or extrinsic/reversible (drugs: β -blockers, CCBs, digoxin; hypothyroidism; vagal tone). Pace only after excluding reversible extrinsic causes.

BOARD TRAP

WRONG: 'symptomatic bradycardia always needs a pacemaker.' First remove reversible causes (rate-slowing drugs, hypothyroidism) before pacing.

7. Thyrotoxicosis → tachy NOT block

WHAT YOU SEE

Tower note: thyrotoxicosis → tachy NOT block.

WHAT IT ENCODES

Hyperthyroidism causes tachyarrhythmias (sinus tachycardia, AF) — not AV block. It's a classic distractor in a bradycardia stem.

BOARD TRAP

WRONG: 'thyrotoxicosis causes high-grade AV block.' It causes tachyarrhythmias; hypothyroidism is the bradycardic one.

8. Coronary split: RCA vs LAD

WHAT YOU SEE

Bottom-left: Inferior/RCA (nodal, transient) vs Anterior/LAD (infranodal, worse → pace).

WHAT IT ENCODES

AV block in inferior MI (RCA) is nodal, transient, atropine-responsive, and usually benign. AV block in anterior MI (LAD) is infranodal from septal necrosis — wide-complex, ominous, and often needs pacing.

BOARD TRAP

WRONG: 'AV block in anterior MI is benign and resolves.' Anterior-MI block reflects extensive infarction and carries a worse prognosis — consider pacing.

9. Bifascicular + syncope → EPS

WHAT YOU SEE

Figure: bifascicular block + syncope → EPS.

WHAT IT ENCODES

Bifascicular block (RBBB + LAFB/LPFB) with syncope suggests intermittent high-grade infranodal block; evaluate with an electrophysiology study or loop recorder, and pace if infranodal block/HV prolongation is found.

BOARD TRAP

WRONG: 'asymptomatic bifascicular block needs prophylactic pacing.' Pace for syncope or documented advanced block, not for the ECG pattern alone.

10. Knight — first sign = AV block (sarcoid)

WHAT YOU SEE

A knight labeled 'first sign = AV block'.

WHAT IT ENCODES

In a young/middle-aged patient with unexplained high-grade AV block, think cardiac SARCOIDOSIS — conduction block is often its first manifestation. Work up with cardiac MRI/FDG-PET; treat with steroids ± device.

BOARD TRAP

WRONG: 'new complete heart block in a 40-year-old is just idiopathic.' Screen for infiltrative disease (sarcoid) and Lyme/giant-cell myocarditis.
